

WO-VSEPR_SIM-63A-ARCHIVE

Hierarchical Sampling and Convergence Testing for High- N Bead Simulation

VSEPR-SIM Development Day 63

Archive Draft

Abstract

This work order archive defines a practical hierarchy for high- N bead-resolved simulation in VSEPR-SIM. Direct all-pairs interaction evaluation scales as $O(N^2)$ and becomes infeasible for million-bead regimes. The document uses octree decomposition as the computational baseline and introduces a dynamic tree sampling method in which local subdivision depth and measurement density adapt to bead concentration, variance, defect concentration, and convergence instability. The purpose is constrained: not to claim atomistic electronic truth from encoded beads, but to test whether bead-level interaction rules converge toward stable macro-scale measurements under increasing sample count and representative-volume scale. Limitations and validation gates are made explicit to prevent overclaiming.

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1 Introduction

Pairwise interaction models scale quadratically:

$$\text{cost}_{\text{all-pairs}} = O(N^2) \quad (1)$$

For $N = 10^6$ beads, direct evaluation implies approximately

$$N(N-1)/2 \approx 5 \times 10^{11} \quad (2)$$

interaction checks per step (order of 10^{12} operations when force kernels and bookkeeping are included).

This is the core bottleneck in large-bead simulation.

1.1 Why high- N is still valuable

Low- N runs reveal local structure and implementation behavior. High- N runs are needed for:

- distribution-level measurements,
- finite-size convergence testing,
- representative-volume behavior,
- macro-property stability under repeated seeds.

Key statement: high- N improves statistical convergence, not microscopic truth by itself.

2 Established Octree Method

In 3D, each cube is subdivided into $2 \times 2 \times 2 = 8$ child cubes. This gives the canonical octree hierarchy.

The force decomposition is:

$$\mathbf{F}_i = \mathbf{F}_{i,\text{near}} + \mathbf{F}_{i,\text{far}} \quad (3)$$

Near-field terms are evaluated explicitly; far-field terms are approximated via aggregate node descriptors.

The Barnes–Hut acceptance condition is:

$$\frac{s}{d} < \theta \quad (4)$$

where:

Symbol	Meaning
s	node size
d	distance from target bead to node center
θ	approximation tolerance

This relation provides the computational backbone: node aggregation is accepted only when geometric separation supports bounded approximation error.

3 VSEPR-SIM High- N Bead Interpretation

To prevent overclaiming, the simulated units are defined as *bead-resolved state carriers*, not guaranteed fully resolved quantum atoms.

A bead state may encode:

$$\{m, \mathbf{r}, q, \epsilon, \sigma, T, \mathbf{x}, \mathbf{v}, \text{state}, \text{material_tag}\} \quad (5)$$

High- N targets improvement in stable measured properties such as:

$$\rho(\mathbf{x}), \phi, D_{\text{eff}}, \text{coordination}, \text{defect fraction}, \text{local variance}, \text{representative-volume stability} \quad (6)$$

These outputs are interpreted as convergence behavior of the encoded model, not direct proof of chemical or electronic correctness.

4 Dynamic Tree Method (Day 63 Contribution)

The standard octree is geometry-driven and mostly static in refinement policy. The dynamic tree method extends this by making local refinement and sampling adaptive to measurement quality.

4.1 Adaptive refinement criteria

Possible split triggers for node ν :

$$n_\nu > n_{\text{max}} \quad (7)$$

$$\sigma_\rho(\nu) > \tau_\rho \quad (8)$$

$$\sigma_P(\nu) > \tau_P \quad (9)$$

$$|\nabla P(\nu)| > \tau_\nabla \quad (10)$$

$$E_{\text{local}}(\nu) > \tau_E \quad (11)$$

where P is a sampled property (porosity, density, coordination, defect fraction, or other tracked field).

4.2 Dynamic behavior rules

Condition	Action
too many beads in node	subdivide
low variation	keep coarse
high defect variance	refine
high boundary/edge error	resample or reject
stable measurement	stop refining
unstable measurement	increase samples

This reframes the method as **adaptive hierarchical sampling for bead-resolved material inference**.

5 Why Not Purely Random Cube Sampling

Random cube probes are useful for analysis,

$$C_k = \text{cube}(\mathbf{x}_k, s_k) \quad (12)$$

but unconstrained random size creates scale bias because $V = s^3$ and large cubes dominate aggregate statistics.

Recommended design:

$$\text{fixed scale bands} + \text{random centers} + \text{convergence reporting} \quad (13)$$

Example scale set:

$$s \in \{5, 10, 25, 50, 100, 250, 500\} \quad (14)$$

For each scale s , report:

$$\mu_P(s), \sigma_P(s), \text{CV}_P(s) \quad (15)$$

with coefficient of variation

$$\text{CV}_P(s) = \frac{\sigma_P(s)}{\mu_P(s)} \quad (16)$$

This measures stability directly rather than smoothing away heterogeneity.

6 Limitations in Goodness Fitting

The method cannot prove that encoded beads are physically correct. It can test whether the encoded bead model converges to stable measured properties under increased sample size and scale.

Limitation	Meaning
bead encoding quality	poor bead rules can converge to stable but unphysical output
interaction law choice	force law determines what is physically represented
sampling scale	too small over-amplifies local noise
large-volume dominance	large samples can hide local critical variation
smoothing risk	smoothing can erase true heterogeneity
parameter explosion	too many fit knobs can become non-identifiable
validation shortage	without empirical anchors, no truth claim is justified

Core sentence: high- N improves convergence of the model, not correctness of the model.

7 Proposed Convergence Gates

Use explicit acceptance gates instead of high-dimensional ad hoc goodness functions. A property P is scale-stable if:

$$\text{CV}_P(s) < \tau \quad (17)$$

and

$$\left| \frac{\mu_P(s_{k+1}) - \mu_P(s_k)}{\mu_P(s_k)} \right| < \epsilon \quad (18)$$

across multiple random seeds.

Example defaults:

$$\tau = 0.05, \quad \epsilon = 0.03, \quad \text{seed_repeat_count} = 3 \quad (19)$$

This yields a conservative and defensible validation layer.

7.1 Gate decision table

Gate result	Interpretation
$CV_P(s) < \tau$ and drift $< \epsilon$ across seeds	Scale-stable for property P at tested bands
$CV_P(s) < \tau$ but drift $\geq \epsilon$	Local variance controlled, but inter-scale consistency unresolved
$CV_P(s) \geq \tau$ and drift $< \epsilon$	Mean appears stable while fluctuations remain too high
$CV_P(s) \geq \tau$ and drift $\geq \epsilon$	Not converged; refine tree, increase samples, or revisit encoding

7.2 Seed-robust pass policy

For each property P , accept only if at least 2/3 seed runs pass both gates on the two largest scale bands. This avoids false positives caused by single-seed luck in heterogeneous regions.

8 Day 63 Framing

Day 63 scope should prioritize hierarchical sampling and convergence testing, not new physics extensions. The octree path supplies scalable structure; the dynamic tree layer supplies adaptive measurement discipline.

8.1 Update-loop doctrine (Day 63+)

Instead of a permanently one-way pipeline

$$\text{property prediction} \rightarrow \text{bead dynamics} \quad (20)$$

the Day 63+ direction is a recursive coupling:

$$S_t \rightarrow P_t \rightarrow D_t \rightarrow S_{t+\Delta t} \quad (21)$$

with

Symbol	Meaning
S_t	bead state at time t (positions, velocities, identities, local structure)
P_t	inferred property field at time t (density, porosity, coordination, transport proxy)
D_t	bead-dynamics operator using inferred properties
$S_{t+\Delta t}$	updated bead state

Formally:

$$P_t = I(S_t) \quad (22)$$

$$S_{t+\Delta t} = D(S_t, P_t) \quad (23)$$

$$P_{t+\Delta t} = I(S_{t+\Delta t}) \quad (24)$$

This is not circular reasoning as validation; it is recursive state–property coupling as dynamics.

8.2 Paper-safe wording and non-claim boundary

In this framework, circularity is not used as validation; it is used as coupling. VSEPR-SIM infers local material descriptors from the bead-resolved state, then bead dynamics uses those descriptors to update the system. The updated state is then reanalyzed, producing a recursive state-property loop. This loop is valuable because many material processes are path-dependent: structure determines properties, while those properties govern the next structural transition.

Claim hygiene:

- Bad claim: the model proves its own properties because the dynamics reproduce them.
- Good claim: the model evolves self-consistent property fields under encoded bead rules.
- Better claim: the framework tests whether recursive bead-property coupling converges, destabilizes, or produces path-dependent material behavior.

Day 63+ doctrine:

Circularity is forbidden as validation, but essential as dynamics.
The loop is not proof. The loop is the model.

Deliverables should therefore emphasize:

- high- N spatial decomposition and traversal behavior,
- scale-band convergence reporting,
- multi-seed stability checks,
- explicit non-claim boundaries.

8.3 Intentional Day 63 bead-model simplicity

For this paper/method, the bead model remains intentionally simple:

1. spherical beads,
2. assigned mass/radius/material tag,
3. optional internal state,
4. local interaction law,
5. sampled property inference,
6. convergence reporting,
7. cleanup and reproducibility pass.

9 Implementation Notes for VSEPR-SIM

A minimal integration path:

1. Add dynamic refinement thresholds to runtime configuration.
2. Record per-node variance and sample stability counters.
3. Export scale-band summaries as structured report tables.
4. Gate pass/fail by (CV, ϵ) constraints across seeds.

Suggested report columns per property and scale:

$$\{s, N_{\text{samples}}, \mu_P, \sigma_P, CV_P, \Delta\mu_{k \rightarrow k+1}, \text{pass/fail}\} \quad (25)$$

9.1 Runtime pseudocode

```
for each seed in seeds:
  build root node from domain bounds
  while refinement_queue not empty:
    node = pop(refinement_queue)
    sample_property(node)
```

```

    if split_trigger(node):
        subdivide(node)
        push(children)
    else:
        commit(node)

for each property P and each scale band s:
    compute mu_P(s), sigma_P(s), CV_P(s)
    compute drift = |mu_P(s[k+1]) - mu_P(s[k])| / mu_P(s[k])
    evaluate convergence gates

aggregate seed pass rates
emit archive report

```

9.2 Minimal archive artifact set

Recommended outputs for reproducibility:

- Node statistics dump: occupancy, split depth, trigger reason.
- Scale-band summary table for each tracked property.
- Seed-level pass/fail ledger with gate values.
- Run manifest: simulation hash, threshold set, and timestamp.

10 Particle Roles and Transient Carrier Dynamics

10.1 Role decomposition

The bead model is extended with multiple particle classes using different update rules, force sensitivities, and event behavior.

Class	Role	Primary behavior
Beads	heavy state carriers	define material structure and long-range field landscape
Transient charged carriers	light/mobile carriers	mediate charge redistribution, local EM effects, capture/emission
Transient neutral carriers	uncharged mobile carriers	ballistic transport with stochastic collision/absorption/scattering

This gives a clean three-level architecture: heavy structural state, mobile charged response, and mobile neutral collision response.

10.2 Beads as heavy structural carriers

Each bead i stores

$$\mathbf{B}_i = (\mathbf{x}_i, \mathbf{v}_i, m_i, r_i, q_i, E_i, s_i) \quad (26)$$

with position, velocity, mass, interaction radius, charge, internal energy state, and material/state label.

Beads provide octree aggregates such as M_{node} , Q_{node} , center-of-mass, and center-of-charge descriptors.

10.3 Transient charged carriers

Define charged carrier state

$$\mathbf{C}_k = (\mathbf{x}_k, \mathbf{v}_k, m_k, q_k, E_k) \quad (27)$$

with light mass, high mobility, and subcycled time updates.

Electrostatic response:

$$\mathbf{F}_k^{\text{EM}} = q_k \mathbf{E}(\mathbf{x}_k) \quad (28)$$

where $\mathbf{E}$ may include bead-octree fields, nearby charged-carrier corrections, external fields, and boundary contributions.

Capture on bead i applies deltas:

$$q_i^{\text{new}} = q_i + q_k, \quad m_i^{\text{new}} = m_i + m_k \quad (29)$$

Mass contribution from electron-like carriers may be neglected if the model is configured for charge-dominant updates.

10.4 Transient neutral carriers

Define neutral carrier state

$$\mathbf{N}_\ell = (\mathbf{x}_\ell, \mathbf{v}_\ell, m_\ell, E_\ell), \quad q_\ell = 0 \quad (30)$$

so electrostatic forcing is disabled:

$$\mathbf{F}_\ell^{\text{EM}} = 0 \quad (31)$$

Neutral motion is modeled as ballistic transport plus stochastic collision/scattering/absorption events with beads. At this scale, default behavior is **not** gravity-tree driven.

10.5 Multi-time-stepping and tree strategy

Use a slow bead step and fast transient substeps:

$$\Delta t_B = 1.0 \text{ fs}, \quad \Delta t_C = 0.01 \text{ fs}, \quad K = \frac{\Delta t_B}{\Delta t_C} = 100 \quad (32)$$

Operational policy:

- rebuild bead octree each bead step,
- use local overlay structures for transient substeps (cell list / bucket grid / local map),
- store event deltas in pending logs,
- apply deltas at bead step boundary (baseline) or propagate deltas up the node path (higher temporal accuracy).

Hybrid trigger:

$$\text{if capture count small} \Rightarrow \text{propagate deltas; else trigger early rebuild.} \quad (33)$$

10.6 Force/effect method by pair type

Interaction	Method	Notes
Bead $\leftrightarrow$ bead bonded	fixed bond list	springs/polymer/structural skeleton
Bead $\leftrightarrow$ bead short-range	neighbor list	LJ/contact/adhesion/damping
Bead $\leftrightarrow$ bead long-range	octree	electrostatic or configured long-range field
Charged carrier $\leftrightarrow$ bead	bead tree + local correction	distant bead charge via tree
Charged carrier $\leftrightarrow$ charged carrier	local neighbor list	avoid default $O(N_C^2)$ unless tiny N_C
Neutral carrier $\leftrightarrow$ bead	stochastic collision model	ballistic + event model
Neutral carrier $\leftrightarrow$ neutral carrier	usually omitted	include only for dense neutral flux
Carrier $\leftrightarrow$ boundary	event policy	escape/reflection/absorption by setup

10.7 Capture, emission, and reaction events

For carrier p entering bead interaction region $\|\mathbf{x}_p - \mathbf{x}_i\| < R_i^{\text{int}}$, evaluate event probability by either rate model

$$P_{\text{capture}} = 1 - \exp(-\lambda_i \Delta t) \quad (34)$$

or path-length model

$$P_{\text{capture}} = 1 - \exp(-\Sigma_i \Delta \ell) \quad (35)$$

then draw $u \sim U(0, 1)$ and accept event if $u < P_{\text{capture}}$.

Charged-capture update:

$$q_i^{\text{new}} = q_i + q_p, \quad m_i^{\text{new}} = m_i + m_p, \quad E_i^{\text{new}} = E_i + E_p + \Delta E_{\text{bind}} \quad (36)$$

Neutral-capture update:

$$q_i^{\text{new}} = q_i, \quad m_i^{\text{new}} = m_i + m_p, \quad E_i^{\text{new}} = E_i + E_p + \Delta E_{\text{reaction}} \quad (37)$$

and optional state transform

$$s_i^{\text{new}} = \mathcal{R}(s_i, p) \quad (38)$$

Emission event model:

$$P_{\text{emit}} = 1 - \exp(-\lambda_{\text{emit}}(E_i, s_i) \Delta t) \quad (39)$$

with conservation updates

$$q_i^{\text{new}} = q_i - q_p, \quad m_i^{\text{new}} = m_i - m_p, \quad E_i^{\text{new}} = E_i - E_p - E_{\text{cost}} \quad (40)$$

and emitted velocity initialization

$$\mathbf{v}_p = v_0 \hat{\mathbf{n}} \quad (41)$$

where $\hat{\mathbf{n}}$ is isotropic or field-biased.

10.8 Simulation loop and logging doctrine

Reference loop:

```

for each bead timestep:
  rebuild bead octree
  rebuild bead neighbor list if needed

  compute bead-bead bonded/short/long-range forces

  for each transient substep:
    update charged carrier forces from bead tree
    update charged local interactions
    update neutral ballistic motion

    check capture, emission, collision, boundary events
    store pending bead mass/charge/energy deltas
    optionally propagate charge deltas up tree

  apply pending bead updates
  integrate bead positions and velocities
  write diagnostics and event log

```

Event records should be separated from state trajectory outputs. State files store evolution; event logs store transformations.

```

{
  "time": 125.04,
  "event": "charged_capture",
  "bead_id": 48291,
  "particle_id": 903,
  "delta_q": -1.0,
  "delta_m": 0.00054858,
  "delta_E": 2.31,
  "position": [4.2, 8.1, 0.5]
}

```

10.9 Naming and claim safety

Section naming should use **Transient Charged and Neutral Carrier Particles**, with electron-like or neutron-like behavior treated as special cases.

This preserves physical caution while retaining the event-driven architecture required for coupled bead-carrier dynamics.

10.10 Signed particle-type assignments (negative codes)

For implementation convenience, transient and special particle classes may use **negative type assignments**. These are *classification codes*, not physical sign assignments for mass or energy.

Type code	Particle class	Notes
+1, +2, ...	Beads (material-resolved)	Primary heavy structural carriers
-1	Charged transient (electron-like)	Fast mobile charge carrier
-2	Charged transient (ion-like)	Optional positive/negative by q field
-3	Neutral transient (neutron-like)	Ballistic + stochastic collision model
-4	Neutral transient (gamma-like packet)	Event/transport packet; no rest-mass update by
-5	Heavy transient (alpha-like packet)	Composite carrier/event token
-6 and below	Reserved special channels	Decay products, diagnostics, boundary-injected p

Recommended rule set:

- Type sign determines class family (positive = bead, negative = transient/special).
- Physical properties remain explicit fields: (m, q, E, s) .
- Neutrality is encoded by $q = 0$, not by type sign alone.
- Gamma-like behavior should be configured as an event/packet model, not interpreted as a bead mass contribution unless explicitly enabled.

This convention simplifies routing and pair-dispatch logic in kernels:

$$\text{if type} > 0 \Rightarrow \text{bead path}; \text{if type} < 0 \Rightarrow \text{transient/special path} \quad (42)$$

while preserving physically meaningful state updates through explicit per-particle fields.

11 Conclusion

WO-VSEPR_SIM-63A is positioned as a methodological archive for scalable inference in bead-resolved simulation. The contribution is practical: combine established octree acceleration with adaptive sampling and convergence gates to decide whether macro-properties are stable under scale and seed variation.

The document intentionally separates: (1) computational convergence claims from (2) physical truth claims.

That separation is required for technically defensible high- N reporting.